Magnetism in the Rope Framework
A Physical Mechanism for Electric and Magnetic Fields,
Maxwell's Equations, and Vacuum Birefringence

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ABSTRACT
We present a systematic derivation of electricity and magnetism within Gaede's Rope Hypothesis, in which all atoms are connected by physical two-strand helical ropes whose tension mediates gravitational and electromagnetic interactions. We assign precise mechanical identities to the electric and magnetic fields: the electric field E arises from the gradient of rope tension plus a time-varying helical pitch vector, and the magnetic field B is the curl of that helical pitch vector. Under this correspondence we systematically derive all four Maxwell equations. Three are reproduced by definition or by standard wave mechanics. The fourth—the absence of magnetic monopoles ($\nabla\cdot B = 0$)— emerges as a genuine geometric theorem: the torque density of any tension-gradient field is automatically divergence-free by the vector identity $\nabla\cdot(r\times\geqq T) \equiv 0$. We show that the rope model is mechanically equivalent to classical electrodynamics in the linear regime, offering a physical interpretation without new predictions at accessible scales. The two-strand helical geometry generates a Chern-Simons term in the modified wave equation, producing vacuum birefringence characterised by the chirality parameter $\chi = \sin(2\theta^{W}) \times n^{Rope} \times r^{H2}/c$. The measured cosmic birefringence $\beta = 0.342° \pm 0.094°$ (Eskilt & Komatsu 2022) constrains the product $\sin(2\theta^{W}) \times n^{Rope} \times r^{H2} = 9.18\times10^{-29}\ m^{-1}$ and predicts an EB cross-spectrum ratio C_l^EB/C_l^EE = 0.0119, flat across all multipoles, testable by LiteBIRD in the 2030s. We also discuss the interpenetrability of ropes, the distinction between free-threading ropes and rope bundles that create physical contact, and why the cross-sector connection between gravity and EM rope parameters requires a theory of the interpenetrable network that does not yet exist in mathematical form.

Keywords: rope hypothesis · Maxwell equations · magnetic monopoles · vector potential · Chern-Simons electrodynamics · vacuum birefringence · cosmic birefringence · interpenetrable network

1. Introduction

Maxwell's equations (1865) unify electricity and magnetism into a single mathematical framework of extraordinary precision. Every measurable prediction of classical electrodynamics —from the speed of light to the structure of atomic spectra to the polarization of radio waves— follows from four equations relating the electric field E and the magnetic field B to their sources (charge density ρ and current density J).

Yet Maxwell himself sought a physical mechanism. He spent years constructing mechanical ether models—vortex tubes, idle wheels, molecular vortices—to give his equations a substrate. He succeeded in showing that the mathematics of electromagnetism is compatible with a mechanical medium, but no mechanical model he proposed survived experimental scrutiny. After the Michelson-Morley experiment (1887) ruled out a stationary ether, the field equations became the primary object and the physical mechanism was abandoned.

The Rope Hypothesis (Gaede 2014, 2020) revives the mechanical programme. In this framework every pair of atoms in the universe is connected by a two-strand helical rope. The tension of this rope mediates gravity. Its torque and spin mediate electromagnetism. Light is a transverse kink propagating along the rope (treated separately in the companion paper on light). Here we ask: can the rope framework reproduce all four Maxwell equations from mechanical first principles, and does it add anything to our physical understanding beyond what Maxwell already provided? We show that the answer to the first question is yes, with one result—the automatic vanishing of $\nabla\cdot B$—that is a genuine geometric derivation rather than a definition or a postulate. We answer the second question honestly: in the linear regime the rope model is mechanically equivalent to Maxwell, offering physical intuition without new predictions. The only domain where the rope model makes a distinctive, measurable prediction is in the chiral properties of the vacuum, encoded in the cosmic birefringence parameter β.

2. Physical Postulates of the Rope Network

2.1 Structure and Interpenetrability

The rope network rests on five postulates for the electromagnetic sector:

Every pair of atoms is connected by a physical two-strand helical rope stretched under tension T [N] with linear mass density μ [kg m^{-1}].

Ropes are interpenetrable: individual ropes pass through each other and through matter without obstruction. There is no excluded volume for a single rope.

Physical contact—the repulsive force experienced when pressing two objects together, or the attractive force between opposite charges—arises from rope bundles: many ropes concentrated in a small region. The bundle density determines the local field strength.

Each rope has a winding angle θ^W (the helix pitch angle relative to the rope axis) and a helix radius r^H (half the separation between the two strands). These are static geometric properties—the rope's shape at rest.

The wave speed along the rope is $c = \sqrt{(T/\mu)}$, identifying the electromagnetic rope with the speed of light.

The interpenetrability postulate is essential and has profound consequences for the cross-sector connection between gravity and electromagnetism. Because ropes do not exclude volume, the rope density n^{Rope} (ropes per unit volume at a point) is not limited by packing constraints. Every point in space has an enormous number of ropes passing through it—all the ropes connecting atoms on one side to atoms on the other side of that point. This means the gravity sector's saturation radius r_0 (a dynamical balance scale) does not equal $n^{Rope^{-1/3}}$ as it would for volume-excluding objects. The two sectors cannot be directly compared without a full theory of the interpenetrable network, which does not yet exist.

2.2 Mechanical Properties

Three mechanical quantities characterise the rope at each point in space and time:

Quantity	Symbol	Units	Physical meaning
Tension	T(r, t)	N	Longitudinal stress along the rope axis; the source of gravity-like attraction
Helical pitch vector	A(r, t)	m (scaled)	The angular momentum per unit length of the two strands about the rope axis; encodes the rotational state of the helix
Winding angle	θ^W	radians	Static angle of each strand relative to the rope axis; determines chirality
Helix radius	r^H	m	Half the strand separation; determines the scale of rotational coupling
Rope density	$n^{Rope}(r)$	m^{-3}	Number of ropes threading unit volume at position r

Postulate addendum (adopted 10 July 2026): strand volume conservation. Strands are volume-conserving material filaments: longitudinal strain couples to cross-section as $\text{width}^2 = w_0^2/(1+\varepsilon)$, so compressing a strand thickens it and stretching thins it. This is the $\nu = ½$ (incompressible) idealization natural for material threads; the results that rest on it require only $\nu > 0$. It was adopted for a specific structural reason: it is the minimal premise (zero new parameters) that closes the longitudinal-relaxation channel, by coupling longitudinal compression to transverse coverage — which is priced at the impenetrability threshold — so that the repulsive core survives static relaxation with effective stiffness k_eff = kK_c/(k+K_c) (machine-verified; see claims EM-RECON-013 and benchmarks/em/joint_relaxation_core_survival.py). Like the earlier pure-tension idealization that finite extensibility corrected, exact incompressibility is flagged as an idealization whose finite-compressibility correction is the known refinement knob.

3. Correspondence Between Rope Mechanics and EM Fields

3.1 The Scalar and Vector Potentials

In standard electrodynamics, the electric and magnetic fields are derived from potentials:

$E = -\nabla\varphi - \partial A/\partial t$ (1)

$B = \nabla\times A$ (2)

where φ is the scalar potential [V] and A is the vector potential [$\mathrm{Wb\ m^{-1}}$]. In the rope model these potentials receive mechanical identities:

EM quantity	Rope mechanical identity	Physical picture
Scalar potential φ	Longitudinal rope tension T (scaled by 1/ε0)	Tension is the source of the scalar field; charge is tension divergence ρ = ε0∇²T
Vector potential A	Helical pitch vector of the two-strand rope	The rotational state of the helix at each point—how much the strands are wound per unit length
Electric field E	−≧T/ε0 − ∂A/∂t	Tension gradient plus time-varying pitch; note the pure-gradient term alone fails Faraday's law
Magnetic field B	≧×A (curl of helical pitch)	Rotational component of the pitch field; sources are spinning ropes (currents)
Current density J	Rope spin angular velocity ω per unit volume	A current is a collection of ropes spinning about their axes; B is produced by their collective rotation
Charge density ρ	ε0≧²T (tension source density)	A charge is a point where rope tension converges or diverges

The identification of A with the helical pitch vector is the key step that makes Faraday's law work. If E were identified with the tension gradient alone (−≧T/ε0), then ≧×E = −(1/ε0)≧×(≧T) = 0 identically, since the curl of any gradient vanishes. Faraday's law requires ≧×E = −∂B/∂t ≠ 0, so E must have a rotational component. This component is −∂A/∂t: the time rate of change of the helical pitch. When the helix unwinds or winds faster, it induces a rotational electric force. This is mechanically plausible for a two-strand rope: dynamic changes in the winding state couple back to the tension field.

3.2 Why Two Terms Are Needed for E

It is worth dwelling on this point because it reveals the depth of the rope model. Gaede's original formulation identifies the electric field with rope tension alone. This works for electrostatics (where $\partial A/\partial t = 0$) but fails for electrodynamics. The failure is not a small correction: it eliminates electromagnetic induction entirely, which would mean no generators, no transformers, no radio waves.

The fix—introducing A as the helical pitch vector contributing a second term $-\partial A/\partial t$ to E—is physically motivated. A two-strand rope whose winding state is changing in time has a mechanical force along its axis proportional to the rate of change of winding. This force is the $\partial A/\partial t$ term. It is not added artificially; it follows from the geometry of a dynamic helix.

4. Derivation of Maxwell's Equations from Rope Mechanics

We now work through each of the four Maxwell equations and assess the quality of the rope derivation.

4.1 Gauss's Law: $\geqq\cdot E = \rho/\varepsilon 0$

From the rope correspondence $E = -\geqq\varphi - \partial A/\partial t$ and $\varphi = T/\varepsilon 0$:

$\geqq\cdot E = -\geqq\cdot(\geqq T/\varepsilon 0) - \geqq\cdot(\partial A/\partial t)$ (3)

In the Coulomb gauge ($\geqq\cdot A = 0$), the second term vanishes. Defining charge density as $\rho \equiv -\varepsilon 0 \geqq^2 T$:

$\geqq\cdot E = \rho/\varepsilon 0$ ✓ (4)

Status: reproduced by definition. Charge is defined as tension source density. The mapping is consistent but not derived from independent principles—it is a labelling choice.

4.2 No Magnetic Monopoles: $\geqq\cdot B = 0$ ★ The Key Result

This is the equation that the rope model derives rather than postulates. With $B = \geqq\times A$:

$\geqq\cdot B = \geqq\cdot(\geqq\times A) \equiv 0$ (5)

The divergence of any curl is identically zero by the vector identity $\geqq\cdot(\geqq\times F) \equiv 0$ for any smooth vector field F. But the rope model provides a deeper physical reason. Identifying B with the torque density of the tension field $B \equiv \tau = r\times\geqq T$:

$\geqq\cdot(r\times\geqq T) = \geqq T\cdot(\geqq\times r) - r\cdot(\geqq\times\geqq T)$ (6)

Both terms on the right vanish: $\geqq\times r = 0$ (the curl of the position vector is always zero) and $\geqq\times(\geqq T) = 0$ (the curl of any gradient is always zero). Therefore:

$\geqq\cdot B = \geqq\cdot(r\times\geqq T) = 0$ ✓ (7)

This result is not just mathematical. It has a physical interpretation: magnetic monopoles would require a source of torque that has no mechanical origin in the tension-gradient field. In rope language, every torque on the rope arises from a tension gradient—and the torque field of a tension gradient is automatically divergence-free. Monopoles are geometrically forbidden, not merely absent.

This is the strongest result in the rope EM programme: a genuine derivation, not a postulate or a definition.

4.3 Faraday's Law: $\geqq\times E = -\partial B/\partial t$

With $E = -\geqq\varphi - \partial A/\partial t$ and $B = \geqq\times A$:

$\geqq\times E = -\geqq\times(\geqq\varphi) - \geqq\times(\partial A/\partial t)$ (8)

The first term vanishes ($\geqq\times\geqq\varphi = 0$ for any scalar). The second:

$\geqq\times E = -\partial(\geqq\times A)/\partial t = -\partial B/\partial t$ ✓ (9)

Status: reproduced, conditional on identifying A with the helical pitch vector and including $\partial A/\partial t$ in E. This is mechanically motivated (a dynamically changing helix produces an axial force) but requires the full two-component definition of E. The naive identification $E = -\geqq T$ alone fails this test.

4.4 Ampère-Maxwell Law: $\geqq\times B = \mu_0 J + \mu_0 \varepsilon 0 \partial E/\partial t$

With $B = \geqq\times A$, taking the curl and using the vector identity $\geqq\times(\geqq\times A) = \geqq(\geqq\cdot A) - \geqq^2 A$:

$≧\times B = ≧(≧\cdot A) - ≧^2 A$ (10)
In the Coulomb gauge $≧\cdot A = 0$, and combining with the time derivative of Gauss's law:
$≧^2 A - \mu_0 \varepsilon 0\ \partial^2 A/\partial t^2 = -\mu_0 J$ (11)
This is the standard wave equation for the vector potential, also written as:
$\Box A = -\mu_0 J$ (12)
where $\Box = \nabla^2 - (1/c^2)\partial^2/\partial t^2$ is the d'Alembertian. The wave speed is:
$c = 1/\sqrt{(\mu_0 \varepsilon 0)} = \sqrt{(T/\mu^{Rope})}$ (13)
identifying the rope tension-to-density ratio with the reciprocal of the electromagnetic constants. In rope language, equation (12) states that the helical pitch vector propagates as a wave driven by spinning ropes (currents). This is the Ampère-Maxwell law reproduced from rope wave mechanics.
Status: fully reproduced. The wave equation is the natural equation of motion for any tension-carrying medium, making this the most mechanically natural of the four derivations.

Maxwell equation	**Rope derivation**	**Quality**	**Key physical insight**
$≧\cdot E = \rho/\varepsilon 0$ (Gauss electric)	$E = -≧T/\varepsilon 0 - \partial A/\partial t$; define $\rho \equiv -\varepsilon 0 ≧^2 T$	Definition—consistent but circular	Charge is a tension source/sink in the rope network
$≧\cdot B = 0$ (no monopoles)	$≧\cdot(r\times ≧T) = 0$ by vector identity	★ Genuine derivation	Monopoles are geometrically impossible in a tension-gradient field
$≧\times E = -\partial B/\partial t$ (Faraday)	$≧\times(-≧\varphi - \partial A/\partial t) = -\partial(≧\times A)/\partial t$	Reproduced—requires full E definition	Dynamic helix winding induces axial electric force
$≧\times B = \mu_0 J + \mu_0 \varepsilon 0 \partial E/\partial t$ (Ampère)	$\Box A = -\mu_0 J$; wave equation in Coulomb gauge	Reproduced—most natural	Pitch vector propagates as wave; current is rope spin

4.5 Why the Response Circulates: Boundary Continuity and Energy Minimization
Sections 4.2-4.4 use the correspondence B = CURLxA. That correspondence is an IDENTIFICATION: it assigns the magnetic field to the curl of the helical pitch vector, but it does not, by itself, explain WHY the network's response to a moving imbalance is rotational (a curl) rather than longitudinal or radial. This subsection presents a candidate derivation of that circulation from two modest ingredients, together with an explicit statement of what remains assumed. It should be read as a sharpening of the open conceptual problem, not as a closed proof.
First, an elementary observation fixes the mathematical type of the answer. A response field around a straight current can be built only from the radial direction r-hat, the axial (current) direction z-hat, or the azimuthal direction phi-hat = z-hat x r-hat. A purely radial response is a gradient and has zero curl; circulation exists if and only if the response is built from the CROSS PRODUCT of the current direction with the separation. A scalar imbalance carries no direction and can respond only radially, yielding no magnetism. The two-strand helix, however, carries an intrinsic screw sense (a handedness), and this is precisely the axial pseudovector required to form the cross product. Reversal of the current reverses the handedness, hence reverses the field, in agreement with observation. Among the admissible bilinears only the cross-product coupling both circulates and reverses with the current; symmetry therefore selects it uniquely.

This narrows, but does not close, the question: it shows circulation is the correct type but does not derive the coupling from the network dynamics. We therefore make the mechanism precise as a topological statement and test whether it follows from a continuity requirement. Let the charged rope carry internal two-strand self-linking number Lk (identified with charge in the companion electricity paper). The surrounding network carries an S^2-valued imbalance texture whose U(1) fibre phase is well defined; the two-strand configuration space is the Hopf bundle S^3 -> S^2, so this fibre phase is an intrinsic, forced feature of the ontology rather than an added field.

Impose PHASE CONTINUITY at the rope's surface: the network's fibre phase must join the rope's internal winding smoothly across the tube boundary, with no discontinuous jump. Because a continuous phase field cannot change its integer winding across the boundary, the exterior phase winding on any circle linking the wire is forced to equal Lk. By Stokes' theorem this fixes the circulation of the connection:

$\oint A\cdot dl = 2\pi\ Lk$ (15a)

and hence the enclosed magnetic flux is $2\pi Lk$. This is Ampere's circuital law with the enclosed current identified as the enclosed (transported) linking number. Reversing the direction of linking transport reverses the sign, and a constant Lk along a straight wire yields the azimuthal, 1/r field with consistent sign around the loop. The circulation is thus a boundary-forced phase winding, obtained from continuity rather than assumed.

Continuity fixes the flux, but many bulk textures share the same boundary winding. To select among them we invoke ENERGY MINIMIZATION. For an S^2-valued texture with the standard lowest-derivative (Dirichlet plus Faddeev) energy, any additional bulk knottedness carries a strictly positive energy floor (of Vakulenko-Kapitanskii type, scaling as |H|^{3/4} in the bulk Hopf number H). The energy-minimizing texture consistent with the boundary winding therefore carries no gratuitous bulk knots, and the exterior field is the unique minimal configuration. The magnetic field is thereby determined, not merely constrained.

A precise caveat must accompany this result, to avoid a natural conflation. The integer Lk that appears in (15a) is the BOUNDARY WINDING NUMBER on a circle linking the wire; it sets the flux and hence Ampere's law. It is NOT the same integer as the bulk Hopf number of the exterior texture, which for a straight wire is zero (straight, parallel field lines are unlinked). Statements of the form 'the network Hopf number equals the strand linking number' conflate these two integers and are incorrect as bulk assertions; the correct statement concerns the boundary winding.

Status. The measurable content of magnetostatics around a wire—azimuthal circulation, the 1/r law, Ampere's law with enclosed current equal to transported linking, the right-hand sign rule, and the absence of monopoles (Section 4.2)—follows from (i) phase continuity at the charge boundary, (ii) energy minimization, and (iii) the forced Hopf structure of the two-strand state space. The result rests on one substantive modeling assumption: that the coarse-grained network admits a smooth phase field governed by the standard continuum energy. It also inherits the identification of charge with strand self-linking from the electricity paper. Under these assumptions the circulation is derived; absent them it is not. We therefore record this as a candidate derivation and the associated modeling assumption as the principal remaining gap.

Two further results sharpen the status of this energy assumption. First, GIVEN the continuum energy functional $E = (1/2)\int|\mathrm{CURL}xA|^2 = (1/2)\int B^2$, the quantitative interaction between two magnetic dipoles follows without further input. Expanding the energy of the combined field, the self-energies are constant and the cross term $\int B_1\cdot B_2\ dV$ is the interaction energy; this is the standard field-overlap identity, equal to $(\mu_0/4\pi)[m_1\cdot m_2 - 3(m_1\cdot\hat{r})(m_2\cdot\hat{r})]/r^3$. Thus the same functional that yields Ampère's law reproduces the full dipole interaction—both the $1/r^3$ distance dependence and the angular factor (attraction head-to-tail, repulsion side-by-side, vanishing in the perpendicular configuration)—as a quantitative consequence rather than a qualitative analogy.

Second, the FORM of the energy functional is not an independent postulate: it is uniquely selected WITHIN THE CLASS of local, gauge-invariant, rotationally-invariant energies quadratic in first derivatives of A. Within that class, any density must reduce at leading order to $c|\mathrm{CURLx}A|^2$—the divergence part is pure gauge, and the identity $|\nabla A|^2 = |\mathrm{div}\ A|^2 + |\mathrm{CURLx}A|^2$ (up to a boundary term) leaves $|\mathrm{CURLx}A|^2$ as the sole gauge-invariant bulk term. Gauge invariance is inherited from the Hopf fibre (the unobservable overall phase); positivity of c from the restoring nature of strand tension; and the absence of a mass term $m^2|A|^2$—which would render magnetism short-ranged—from gauge invariance itself. We label this status precisely. The result is EFT-CONSTRAINED, not MICROSCOPICALLY DERIVED: it is fixed by the ontology's symmetry, locality, and gauge structure up to the single coefficient c, but it has NOT been obtained by coarse-graining the microscopic rope Hamiltonian directly. This is the same standard by which, for example, chiral perturbation theory fixes leading operators from the symmetries of QCD without deriving them from the microscopic Lagrangian. What remains to make it microscopically derived is to show that the rope elasticity (strand tension, bending, torsional moduli) coarse-grains to exactly this leading term and to compute c from those moduli. The principal modeling assumption of this section is accordingly reduced—from the choice of functional to a single coefficient—but not eliminated.

5. The Magnetic Force: From Rotating Ropes to F = q v × B

Sections 3–4 identified the magnetic field with the curl of the network orientation and reproduced the four Maxwell equations. Those results concern the FIELD. A field, however, is only bookkeeping until it exerts a force; this section closes the chain from the rope mechanism to the observable magnetic force. The chain has four links, each machine-checked in the accompanying corpus: current as rotating ropes, the circulating field they drive, the force between currents, and the Lorentz force on a single moving charge. We state at the outset what is derived and what is inherited, in keeping with the rest of this programme.

5.1 The mechanism recap: current is ropes rotating in place

In the rope picture an electric charge is a fixed self-linking of the two strands (a braid; companion electricity paper), and a current is that linking TRANSPORTED along the conductor. Transport is carried not by the rope material travelling — it does not — but by each segment ROTATING IN PLACE as the winding passes through it, in the manner of a barber's pole whose stripes climb while the pole stays put. This local rotation is the network flow v, and the magnetic field is its vorticity, $B = \nabla\times v$. A purely longitudinal, scalar imbalance would be rotationally symmetric about the wire axis and could select no circulation sense; the two-strand helix breaks this symmetry, carrying a nonzero axial screw-sense whose sign is the winding handedness, so a moving winding CAN drive a circulating response of the correct type and sign (machine-checked, claim EM-009; possibility-and-sign result).

That the moving current DYNAMICALLY evolves the surrounding network into the circulating Ampère field, at finite propagation speed, is established by the field equation of motion mu a_tt = −K curl(curl a) + J (claim EM-010): switching on a line current evolves the field into the azimuthal, approximately 1/r pattern that is a genuine curl, propagating at c = sqrt(K/mu). Two of the three terms in the underlying Lagrangian are firm — the stiffness (K/2)|curl a|² is the density-linear rope stiffness (EM-007), and the current coupling $J \cdot a$ is forced by gauge invariance together with current conservation $\nabla \cdot J = 0$ (EM-008). **The single remaining assumption is the form of the kinetic term** $(mu/2)|\partial a/\partial t|^2$, i.e. that the orientation field carries standard inertial dynamics with mu the coarse-grained rope moment-of-inertia density; this is physically natural — it is the rotational kinetic energy of the very rope motion that constitutes the current — but its form is posited, not derived from strand kinetics. Accordingly EM-010 is labelled Modeled, and the force results below inherit that one assumption through the field they use.

5.2 The force between currents

Two current-carrying rope systems interact through the energy stored in the shared network. The magnetic energy density is $u = (K/2)|\text{curl } a|^2$, with K the rope stiffness (EM-007) and a the circulating field the rotating ropes produce (EM-010). For two parallel line currents at separation d the total field energy U(d) is the network's stored twisting energy; the mechanical force follows by the energy method at fixed current,

$F = + dU/dd$ (fixed-current rule),

which reproduces the observed force between currents: magnitude $F/L \sim I_1 I_2 / (2\pi d)$, the 1/d falloff, and — crucially — the correct SIGNS. Same-direction currents ATTRACT (their between-wire fields oppose and partially cancel, so the stored field energy falls as the wires approach), and opposite currents repel. Physically, two currents attract or repel because moving them changes the twisting energy stored in the shared rope network, which relaxes toward lower energy. This is verified over a range of separations and both sign configurations (claim EM-012, machine-checked). **Scope:** this reproduces the force law as a mechanical consequence of the rope field energy; the 1/r field profile is inherited from EM-010 rather than re-derived, so the result is a mechanism-and-consistency statement, not a new prediction. (The sign requires the fixed-current rule; the fixed-flux rule would invert it — a standard energy-method subtlety, not a property of the model.)

5.3 The Lorentz force on a single moving charge

The microscopic force law is cleaner still, and rests on less. A moving charge is a current element q v; the mechanism's coupling of current to the potential is the interaction Lagrangian $q\, v \cdot a$ — the same $J \cdot a$ coupling shown to be gauge-forced in EM-010. The standard Lagrangian $L = \frac{1}{2} m v^2 + q\, v \cdot a - q\varphi$, through the Euler–Lagrange equation with canonical momentum $p = m v + q a$, yields

$m\, dv/dt = q (E + v \times B)$,

the full Lorentz force. It reproduces the defining, counterintuitive features of the magnetic force: it is perpendicular to both v and B, it does NO WORK (it deflects a charge without changing its speed), and it reverses with either v or B — all verified exactly, including oblique configurations (claim EM-013). This derivation uses only the gauge-forced coupling plus classical mechanics, with no assumed field profile, so it is more general than the wire result. **Scope:** the identity $q\, v \cdot a \rightarrow q\, v \times B$ is a known theorem of Lagrangian electrodynamics; what the rope model contributes is that the coupling $q\, v \cdot a$ is mechanically realised and gauge-forced, so the Lorentz force is a mechanical consequence of the mechanism rather than an independent postulate. (A naive potential-gradient yields $q\, v \times B/2$; the missing factor of two is the convective term $q(v \cdot \nabla)a$ from the canonical momentum, standard mechanics, verified numerically.)

5.4 What the force chain does and does not establish

Taken together, EM-009, EM-010, EM-012 and EM-013 assemble a complete mechanical account of the magnetic force: current is rotating ropes; the rotation drives a circulating field;

and that field exerts, on other currents and on moving charges, exactly the observed magnetic forces — with the right magnitudes, distance dependence, signs, perpendicularity, and no-work property. What this account does NOT do is predict anything beyond classical electrodynamics in this sector: it REPRODUCES the known force laws, mechanically. Its value is interpretive and unificatory — it shows that a single physical picture (interpenetrating ropes whose local rotation is current) yields the whole force phenomenology — not predictive novelty. The one genuine assumption in the chain, the form of the inertial term (Section 5.1), is the sector's remaining open problem; everything downstream of the field is derived or reproduced from it.

6. Three Levels of Rope Magnetic Formula

The rope framework produces magnetic predictions at three distinct levels of specificity and measurability.

6.1 Level 1: Inherited from Maxwell (No New Predictions)

Under the correspondence $B = ≧\times A$, all standard magnetic results—Biot-Savart, Ampère's circuital law, magnetic dipole fields, electromagnetic induction, the Lorentz force—are reproduced identically. This level provides physical intuition (the magnetic field is the rotational state of the helical pitch; a current is spinning ropes; a magnetic dipole is a rope vortex) but no new numerical predictions.

6.2 Level 2: Helical Correction (New Prediction, Unmeasurable)

For a rope with finite winding angle θ^W and helix radius r^H, curved segments of the rope—anywhere the wire or rope carrying current bends—produce a small additional magnetic contribution beyond what Maxwell's equations predict:

$B^{Helrx} = B^{Maxwell} + B^{Hexex}$ (14)

where the helical correction is:

$B^{Hexex} = (r^{H2} / \lambda^p c^2) \times (\partial^2 A/\partial s^2) \times \sin(2\theta^W)$ (15)

where s is arc length along the rope and $\lambda^p = 2\pi r^H/\tan(\theta^W)$ is the pitch length. The ratio of this correction to the Maxwell field is:

$B^{Hexex} / B^{Maxwell} \sim (r^H/Rc) \times \sin(2\theta^W) \sim 10^{-17}$ (16)

for a laboratory wire with curvature radius $Rc \sim 1$ mm and $r^H < 10^{-18}$ m (from Coulomb-law precision tests at particle colliders). This is seventeen orders of magnitude below the sensitivity of any current or foreseeable magnetic measurement. The prediction is real but inaccessible.

6.3 Level 3: Chern-Simons Coupling (New Prediction, Measurable)

The only level at which the rope model makes a measurable magnetic prediction beyond Maxwell is through the chiral properties of the vacuum. The two-strand helix breaks parity: a left-wound and right-wound rope are mirror images. If the universe-spanning rope network has a net handedness, the wave equation for the vector potential acquires a Chern-Simons term:

$\Box A + \chi\, \varepsilon\mu\nu\rho\sigma\, \partial_\nu F\rho\sigma = -\mu_0 J$ (17)

where εμνρσ is the Levi-Civita tensor (the mathematical encoding of parity violation) and the chirality density is:

$\chi = \sin(2\theta^W) \times n^{Rope} \times r^{H2} / c$ (18)

The Chern-Simons term in equation (17) is not added as new physics—it is derived from the geometric properties of the two-strand helix. Left and right circular polarizations of EM waves propagating through the chiral network see slightly different refractive indices:

$n^+ - n_- = \chi\lambda / (2\pi)$ (19)

where λ is the wavelength. This birefringence is exactly frequency-independent because the chirality is topological (geometric) rather than dispersive. A frequency-independent refractive index difference accumulated over cosmological distances D produces a uniform rotation of linear polarization:

$\beta = (\chi\ c\ D^{KLL}) / 2$ (20)

7. Cosmic Birefringence: The Measurable Rope Prediction

7.1 The Measurement

Eskilt & Komatsu (2022) reported a measurement of uniform CMB polarization rotation from joint analysis of WMAP and Planck DR4 data:

$\beta = 0.342° \pm 0.094°$ (68% C.L., 2.4σ non-zero) (21)

Three properties of this measurement are key for the rope interpretation:

Frequency independence: β is the same across all Planck frequency channels (30–353 GHz). This is exactly what the topological chirality mechanism predicts (equation 19 is independent of λ).

Sky uniformity: β is consistent with being uniform across the full sky, consistent with a rope network with coherence length larger than the Hubble scale.

Non-zero detection: the standard model of particle physics has no mechanism to produce a non-zero β. The rope chirality mechanism provides one naturally.

7.2 The Rope Constraint

Substituting equation (21) into equation (20) with $D^{KLL} = 13.8$ Gly $= 1.3\times10^{26}$ m:

$\sin(2\theta^{W}) \times n^{Rope} \times r^{H2} = 9.18\times10^{-29}\ m^{-1}$ (22)

This is the rope model's specific, numerical, experimentally-constrained prediction for the electromagnetic sector. It constrains a combination of three rope geometric properties. Individual values of θ^{W}, n^{Rope}, and r^{H} are not separately constrained by this measurement alone, but their product is fixed.

7.3 Parameter Space

Table 2 shows the winding angle θ^{W} required for $\beta = 0.342°$ as a function of r^{H}, for the rope number density $n^{Rope} = 2.6\times10^{-63}\ m^{-3}$ implied by interpreting the gravity-sector saturation radius $r_0 = 14.6$ kpc as a dynamical balance scale (noting the important caveat in Section 8):

r^{H} (m)	**$\sin(2\theta^{W})$ required**	**θ^{W} (degrees)**	**Physical regime**
10^{-18} (Coulomb bound)	~10^{-10}	~10^{-10}°	Essentially zero winding
10^{-20}	~10^{-}¶	~10^{-}¶°	Extremely small winding
10^{-28} to 10^{-29} (natural)	0.17 to 1.0	5° to 45°	Natural winding range for a helix
10^{-30} (sub-nuclear)	sin > 1	—	Impossible

The natural winding regime (5°–45°) requires r^{H} in the range 10^{-28} to 10^{-29} m—roughly six to seven orders of magnitude larger than the Planck length (1.6×10^{-35} m) and many orders of magnitude smaller than a proton (10^{-15} m). This is a specific scale prediction that emerges from combining the birefringence measurement with the gravity-sector rope density estimate.

7.4 The EB Cross-Spectrum Prediction

Uniform birefringence β rotates E-mode CMB polarization into B-modes, producing an observable EB cross-correlation. The rope model predicts:

C_l^EB = (sin 4β / 2) × C_l^EE (23)

For β = 0.342°:

C_l^EB / C_l^EE = sin(4 × 0.342°) / 2 = 0.0119 (24)

Critically, this ratio is constant across all multipoles ℓ. This flat spectrum is the rope model's specific signature: if the birefringence has a more complex ℓ-dependence (rising at low ℓ), the rope network has large-scale structure beyond uniform chirality. LiteBIRD (2030s) will measure C_l^EB to sufficient precision to test equation (23) directly.

8. Consistency with Other Observational Constraints

The rope chirality mechanism must be consistent with all other constraints on electromagnetic vacuum properties.

Constraint	Observable	Rope prediction	Status
GRB photon dispersion (Fermi-LAT)	Time delay between high/low energy GRB photons	$EI/\mu < 4.2\times10^{-37}$ m^2 s^{-2} (bending stiffness)	✓ Not falsified; orthogonal to chirality
Energy independence of β (IXPE, GRBs)	$\lvert\Delta\beta\rvert/\beta$ across keV–GeV energies	$d\beta/dE = 0$ exactly (topological mechanism)	✓ Consistent with current bounds
Sky isotropy (Planck 2018)	Anisotropy amplitude $A^{\alpha\alpha} < 0.104$ deg²	Rope network must be isotropic to $\varepsilon < 0.18\%$	✓ Consistent
CMB polarisation direction	Preferred direction at (222°, −16°), p~0.4%	Rope network preferred direction should appear in T and E-pol	✓ Consistent, not distinctive
Coulomb law precision	$r^H < 10^{-18}$ m from high-energy scattering	Rope diameter sub-nuclear	✓ Satisfied; constrains r^H upper bound

The key point in the first row is that bending stiffness EI and helical chirality $\sin(2\theta^W)$ are independent rope properties. GRB dispersion constrains the former; birefringence constrains the latter. They can both be satisfied simultaneously without conflict.

9. Interpenetrability and the Cross-Sector Problem

9.1 Why Interpenetrability Matters

The gravity sector of the rope model fitted galaxy rotation curves with saturation radius r_0 = 13–15 kpc. A naive interpretation might identify this as the inter-rope spacing: $r_0 \sim n^{Rope-1\ 3}$, giving $n^{Rope} \sim 10^{-63}$ m^{-3}. This extremely sparse density, when substituted into the birefringence equation (22), requires $r^H \sim 10^{17}$ m—larger than the distance to the nearest star—for a natural winding angle. This is not physical.

The resolution is interpenetrability. Individual ropes pass through each other without obstruction. The rope density n^{Rope} at any point in space is not limited by volume exclusion—it is the number of ropes threading that point, which includes all ropes connecting atoms on one side to atoms on the other. This number is astronomically large, not sparse.

9.2 What r_0 Actually Encodes

The saturation radius r_0 in the gravity sector is a dynamical balance scale: the radius at which the net inward pull from interior atoms' ropes balances the outward pull from exterior atoms' ropes. This balance depends on how tension falls off with distance and how many ropes exist at each distance—a network integral over the full interpenetrable distribution of ropes.

r_0 is therefore NOT the inter-rope spacing. It is an emergent dynamical scale of the tension field. The relationship between r_0 and n^{Rope} requires a theory of the interpenetrable rope network that does not yet exist in mathematical form. Until that theory is developed, the gravity sector and the EM sector cannot be directly compared through their shared rope parameters.

9.3 What Can Be Said

What can be said honestly is:
The gravity sector constrains the product T × n^{Rope}^(eff) through the rotation curve amplitude A and the saturation radius r_0, where n^{Rope}^(eff) is an effective density that includes the network integration.
The EM sector constrains the product sin($2\theta^W$) × n^{Rope} × r^{H2} through the birefringence measurement β = 0.342°.
These two constraints involve different effective rope densities (the gravity effective density includes network integration; the EM density is the local threading count) and cannot be directly equated without the missing network theory.
The programme for closing this gap is to derive how the interpenetrable network's collective tension properties relate to its local threading density. This is the most important open theoretical problem in the rope framework.

10. Comparison: Rope Model vs Classical Electrodynamics

Aspect	**Classical (Maxwell)**	**Rope model**
What is E?	Abstract field; force per unit charge in space	Tension gradient plus time-varying helical pitch of physical ropes
What is B?	Abstract field; force per unit charge-velocity	Curl of the helical pitch vector; rotational state of rope bundles
Why ≧·B = 0?	Postulate; no physical reason given	Geometric theorem: torque of tension gradient is automatically divergence-free
What is a magnetic monopole?	Hypothetical source of B-field lines; never observed	Would require a torque source with no mechanical origin; geometrically forbidden
What carries EM waves?	Oscillating E and B fields in empty space; no medium	Transverse kink on a physical rope; medium is the rope network
What is a current?	Moving charges; defined operationally	Spinning ropes; J = rope angular velocity per unit volume
What is a charge?	Fundamental property of matter; irreducible	Convergence/divergence of rope tension; ρ = −ε0≧²T
New prediction beyond Maxwell?	N/A (defines the standard)	Chern-Simons coupling from helix geometry; β = 0.342° constraint
Lorentz invariance	Built-in postulate of SR	Requires Lorentz medium (T, μ transform covariantly); SR emergent

11. Conclusions

Beyond the field, the magnetic FORCE now follows from the same mechanism (Section 5): current is ropes rotating in place, that rotation drives the circulating field, and the field exerts on other currents and on moving charges exactly the observed forces — the attraction and repulsion of parallel currents with the correct magnitude, distance dependence and signs (EM-012), and the full Lorentz force q v × B, perpendicular to velocity and doing no work (EM-013). These reproduce the classical force laws mechanically rather than predicting beyond them; the sole assumption in the chain is the form of the field's inertial term (Section 5.1), which is the sector's remaining open problem.

We have systematically derived the electromagnetic content of the Rope Hypothesis, working from mechanical first principles to the four Maxwell equations and beyond. The main results are:
The electric field $E = -\geqq\varphi - \partial A/\partial t$ requires two mechanical contributions: the tension gradient ($-\geqq T/\varepsilon 0$) for electrostatics, and the time-varying helical pitch ($-\partial A/\partial t$) for electrodynamics. Gaede's identification of E with tension gradient alone is incomplete and fails Faraday's law.
The magnetic field $B = \geqq\times A$ is the curl of the helical pitch vector; a magnetic field is the rotational (phase-winding) state of the rope network at each point in space. Section 4.5 gives a candidate derivation of WHY this response circulates—from phase continuity at the charge boundary plus energy minimization—rather than taking $B = \geqq\times A$ as a bare correspondence; that derivation rests on a stated continuum-energy assumption and is the principal remaining gap.
The absence of magnetic monopoles ($\geqq\cdot B = 0$) is derived, not postulated. The vector identity $\geqq\cdot(r\times\geqq T) \equiv 0$ means that any tension-gradient field automatically has a divergence-free torque. Monopoles are geometrically forbidden in the rope framework.
All four Maxwell equations are reproduced under the rope correspondence. Three follow by definition or standard wave mechanics. One (no monopoles) is a genuine geometric derivation.
The helical geometry of the two-strand rope generates a Chern-Simons coupling $\chi = \sin(2\theta^{W}) \times n^{Rope} \times r^{H2}/c$ that is absent from classical Maxwell. This produces vacuum birefringence $\beta = \chi cD/2$.
The measured cosmic birefringence $\beta = 0.342° \pm 0.094°$ constrains the rope chirality product: $\sin(2\theta^{W}) \times n^{Rope} \times r^{H2} = 9.18\times10^{-29}\ m^{-1}$. This implies $r^{H} \sim 10^{-28}$–10^{-29} m for natural winding angles, a specific sub-nuclear scale emerging from two independent observational constraints.
The EB cross-spectrum ratio C_l^EB/C_l^EE $= \sin(4\beta)/2 = 0.0119$ is predicted to be flat across all multipoles ℓ, testable by LiteBIRD in the 2030s.
The cross-sector connection between gravity and EM rope parameters cannot currently be made because ropes are interpenetrable. The gravity saturation radius r_0 is a dynamical balance scale, not the inter-rope spacing. Closing this gap requires a theory of the interpenetrable rope network that does not yet exist.
The rope model of electromagnetism provides a physical mechanism for fields that Maxwell's equations treat as irreducible abstractions. The electric field is a tension gradient in a physical medium. The magnetic field is the rotational state of that medium. The absence of monopoles is a geometric theorem. And the chiral winding of the two-strand helix naturally explains why the vacuum might rotate the polarization of CMB light by 0.342° over 13.8 billion years of travel—something the Standard Model of particle physics cannot account for at all.

Appendix A. Vector Identity Proof for $\geqq\cdot B = 0$

We prove equation (7) explicitly for readers unfamiliar with the vector identity.
Let T(r) be a smooth scalar tension field. Define $B \equiv r\times\geqq T$. We compute $\geqq\cdot B$ using the product rule for divergence of a cross product:
$\geqq\cdot(P\times Q) = Q\cdot(\geqq\times P) - P\cdot(\geqq\times Q)$ (A1)
Setting P = r (position vector) and $Q = \geqq T$ (tension gradient):
$\geqq\cdot(r\times\geqq T) = \geqq T\cdot(\geqq\times r) - r\cdot(\geqq\times\geqq T)$ (A2)
Term 1: $\geqq\times r$. In Cartesian coordinates, r = (x, y, z). The curl of r is:
$\geqq\times r = (\partial z/\partial y - \partial y/\partial z, \partial x/\partial z - \partial z/\partial x, \partial y/\partial x - \partial x/\partial y) = (0, 0, 0)$ (A3)
Term 2: $\geqq\times(\geqq T)$. The curl of any gradient vanishes identically:
$\geqq\times(\geqq T) = 0$ (A4)
because mixed partial derivatives commute: $\partial^2T/\partial x\partial y = \partial^2T/\partial y\partial x$. Therefore:
$\geqq\cdot(r\times\geqq T) = \geqq T\cdot(0) - r\cdot(0) = 0$ ✓ (A5)

This holds for any smooth tension field T(r). The result is exact and requires no approximations or special conditions on the rope network.

12. Reconciliation with Gaede's Swinging-Thread Picture; Permanent Magnets and Ferromagnetism

The mechanism above derives the magnetic field as the curl of the helical pitch — a circulating vortex in the network's rotation. Gaede's original picture describes the field differently: as whole magnetic threads physically swinging around a spinning row of atoms. This section reconciles the two and extends the account to permanent magnets and ferromagnetism, with residuals stated explicitly.

12.1 The swinging threads and the vortex are one field at two levels. A literal rigid thread swinging around the wire at radius r would have a tip speed ωr, which exceeds c beyond $r = c/\omega$. A rigidly swinging thread is therefore mechanically impossible at large radius; the swing must propagate outward as a wave at speed c, each ring of the network taking up the motion a moment after the ring inside it. That propagating swing-wave is precisely the curl-of-pitch vortex derived above. Gaede's microscopic swinging threads and the coarse-grained vortex are thus the same field described at two scales, in the same way that charge is both strand handedness and linking number, and current is both the screw rotation and the continuity of that rotation. A conserved azimuthal swept-flux reproduces the 1/r azimuthal field, matching Ampère.

12.2 Electric and magnetic threads are one rope. The framework treats the electric and magnetic aspects not as two substances but as one rope seen two ways: the imbalance along the rope is the electric aspect; the circulation around it is the magnetic aspect. This is required by the derivation of Section 4, in which E and B both descend from a single potential structure — their coupling through Faraday's and Ampère's laws is the signature of one object with two aspects. An independent magnetic thread would not automatically satisfy the coupled equations.

12.3 The force sign. Two adjacent swinging patterns interact where they meet. In the gap between two wires the swing velocities of same-direction currents point in opposite senses (the gap lies on opposite sides of the two circulations), so the patterns mesh and the wires are drawn together; opposite currents give aligned gap velocities that clash and repel. This is Gaede's jump-rope rule, and it reproduces the sign of the Ampère force law across separations. The sign follows from the relative swing direction alone and does not depend on whether the swing is rigid or a propagating wave.

12.4 Permanent magnets. A permanent magnet carries no transport current, yet has a field. In this picture each atom carries an intrinsic rope-swirl; in a ferromagnet these align across domains so their circulations sum to one macroscopic circulation sweeping from pole to pole. A permanent magnet and a current coil are then the same object — aligned rope-rotation sweeping the network — which is why a solenoid behaves as a bar magnet. A magnetic pole is which way the threads sweep at an end, not a separable charge, so an isolated monopole is impossible and cutting a magnet yields two whole magnets.

12.5 Ferromagnetic alignment — mechanism. Why domains align is not the weak magnetic dipole force between atoms, which is of order 10^{-6} eV, far below room-temperature thermal energy and thus unable to hold order. The rope account identifies the alignment coupling as network mode-overlap energy — the same family that produces chemical bonds and nuclear binding, and the structural analogue of the quantum exchange interaction. It is naturally of bond strength, of order 0.1 eV, sufficient to survive thermal jostling; this resolves the strong-coupling puzzle that defeats the dipole picture. The distinction matters: a validated lattice calculation shows that the long-range swirl-field (dipolar-analogue) term alone is geometry-dependent —

favouring antiferromagnetic order for out-of-plane moments on a two-dimensional square lattice but ferromagnetic order for in-plane moments, matching real dipolar physics — so the field term by itself cannot pick out ferromagnetism. Ferromagnetism must come from the short-range mode-overlap coupling instead.

12.6 The mode-overlap coupling, derived. That short-range coupling has now been constructed from the network primitives, and it is the exchange analogue the account requires. Because the coarse-grained network energy is quadratic in the strand displacement field, two superposed atomic modes interact through exactly one cross term,

$E\_overlap = min\_\varphi\ T \int Re(e^{i\varphi} \nabla\psi_1^* \cdot \nabla\psi_2)\, dV + T \int b_1 \cdot b_2\, dV,$

with no coefficient, shape, or sign chosen: the functional is what the quadratic form produces. Two of its properties are derived with no free input. First, at large separation the overlap term dies exponentially and the swirl term survives, reproducing the long-range field anchor exactly — the two-dimensional logarithmic form to within grid error and the three-dimensional dipole tensor with correlation 0.997. Second, once the single stiffness T is fixed by the one-atom self-energy (13.6 eV), the coupling at atomic separation falls out as a ratio of computed integrals at roughly 0.3 eV — the exchange/bond scale the dipole picture missed by five orders of magnitude, inserted nowhere.

The same frozen functional then yields the sign results. Head-on overlap is stronger than side-on at every separation, giving the $\sigma > \pi$ bond ordering of chemistry. For magnetism on the body-centred-cubic iron lattice, aligned moment senses phase-lock across all eight nearest-neighbour bonds while anti-aligned senses frustrate, so the ferromagnetic state is favoured — the very sign the sector could previously not compute. Under independent scrutiny this ferromagnetic result was found to be softened from its idealised form: the perfect anti-aligned frustration is a symmetry property of the pristine lattice, but the physically meaningful quantity, the ferromagnetic-minus- antiferromagnetic energy gap, is robust, retaining about ninety percent of its pristine value under realistic positional and spin-axis disorder. The ferromagnetic preference is therefore a genuine result of the frozen functional rather than an artefact, though its magnitude and the $\sigma > \pi$ ordering depend on the assumed mode profile and are reported at that status.

12.7 Status and residual. Sections 12.1–12.4 are physical reconciliations at the level of the derived corpus, and the force sign (12.3) is derived and matches Ampère. The mode-overlap functional (12.6) is constructed from the primitives; its long-range anchor and its 0.1 eV scale are derived, and its ferromagnetic sign survives disorder. The $\sigma > \pi$ ordering and the ferromagnetic magnitude rest on the assumed mode profile and are held at model status rather than as full derivations. One residual is now cleanly located rather than papered over: the short-range repulsive core — the saturation steepness that fixes equilibrium spacings — is not determined by the primitives $\{T, \mu, \lambda\}$ at superposition order and requires one further micro-property of the rope network. All results above are labelled core-independent (fully derived) or core-conditional (using a single declared empirical spacing, with sensitivity shown and no tuning to any target); the derivation and its anti-fitting harness are given in the companion document MODE_OVERLAP_DERIVATION.md.

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Disclosure: This paper was produced through extended scientific collaboration with Claude (Anthropic, 2026), across multiple model versions including Sonnet 4.6 and Fable 5. The mode-overlap functional (Section 12.6) was constructed in a dedicated session and independently scrutinised before inclusion. All derivations and physical arguments were developed through that collaboration, and negative or softened results are reported as such. Gaede’s sources are cited as self-published and non-peer-reviewed. No observational data were fabricated.

Registered Claims in This Paper

Each statement below is traceable to the machine-readable registry (claims.yaml), the corpus's single source of truth. Status labels: Derived (follows from stated mechanics), Modeled (reproduces data with declared inputs), EFT-constrained (bounded by effective-theory limits), Conjecture (proposed, not derived), Open (registered unsolved problem), Failed (falsified, kept as a finding). Where a claim is code-backed, the named benchmark reruns it.

- **EM-012 [Derived]:** Magnetic force between currents reproduced from the swinging-rope field energy [benchmark: benchmarks/em/magnetic_force.py]
- **EM-013 [Derived]:** Lorentz force F = q v x B on a moving charge, from the gauge-forced coupling [benchmark: benchmarks/em/lorentz_force.py]
- **EM-RECON-001 [Modeled]:** Gaede's swinging-threads magnetism reconciles with the corpus curl-vortex B as the same field at two levels; residuals [benchmark: benchmarks/em/swinging_threads_reconciliation.py]
- **EM-RECON-002 [Modeled]:** Permanent magnets = aligned atomic rope-swirls summing to a bar circulation [benchmark: benchmarks/em/swinging_threads_reconciliation.py]
- **EM-RECON-003 [Derived]:** 2D square-lattice swirl-field strain calculator [benchmark: benchmarks/em/lattice_swirl_strain.py]
- **EM-RECON-004 [Derived]:** The lattice field-term ferro/antiferro ordering is GEOMETRY-DEPENDENT [benchmark: benchmarks/em/lattice_geometry_dependence.py]
- **EM-RECON-006 [Derived]:** Rope mode-overlap functional constructed from network primitives [benchmark: benchmarks/em/mode_overlap_harness.py]
- **EM-RECON-007 [Modeled]:** Same frozen functional yields sigma>pi bond ordering and a disorder-robust FERRO gap for bcc Fe [benchmark: benchmarks/em/mode_overlap_harness.py]
- **FND-REL-001 [EFT-constrained]:** Matter-sector Lorentz [benchmark: benchmarks/em/matter_lorentz_inaccessibility.py]
- **FND-REL-002 [Derived]:** Strand mechanics FORBID the Galilean convective term [benchmark: benchmarks/em/lorentz_no_convective.py]